

Study on the citrus greening disease: Current challenges and novel therapies

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ABSTRACT

The unprecedented worldwide spread of the Citrus greening disorder, called Huanglongbing (HLB), has urged researchers for rapid interventions. HLB poses a considerable threat to global citriculture owing to its devastating impact on citrus species. This disease is caused by *Candidatus Liberibacter* species (CLs), primarily transferred through psyllid insects, such as *Trioza erytreae* and *Diaphorina citri*. It results in phloem malfunction, root decline, and altered plant source-sink relationships, leading to a deficient plant with minimal yield before it dies. Thus, many various techniques have been employed to eliminate HLB and control vector populations through the application of insecticides and antimicrobials. The latter have evidenced short-term efficiency. While nucleic acid-based analyses and symptom-based identification of the disease have been used for detection, they suffer from limitations such as false negatives, complex sample preparation, and high costs. To address these challenges, secreted protein-based biomarkers offer a promising solution for accurate, rapid, and cost-effective disease detection. This paper presents an overview of HLB symptoms in citrus plants, including leaf and fruit symptoms, as well as whole tree symptoms. The differentiation between HLB symptoms and those of nutrient deficiencies is discussed, emphasizing the importance of precise identification for effective disease management. The elusive nature of CLs and the challenges in culturing them in axenic cultures have hindered the understanding of their pathogenic mechanisms. However, genome sequencing has provided insights into CLs strains' metabolic traits and potential virulence factors. Efforts to identify potential host target genes for resistance are discussed, and a high-throughput antimicrobial testing method using Citrus hairy roots is introduced as a promising tool for rapid assessment of potential treatments. This review summarizes current challenges and novel therapies for HLB disease. It highlights the urgency of developing accurate and efficient detection methods and identifying the complex relations between CLs and their host plants. Transgenic citrus in conjunction with secreted protein-based biomarkers and innovative testing methodologies could revolutionize HLB management strategies toward achieving a sustainable citrus cultivation. It offers more reliable and practical solutions to combat this devastating disease and safeguard the global citriculture industry.

1. Introduction

Citrus fruits have always received significant attention from consumers worldwide for their health-enhancing phytonutrients. They are cultivated in over 140 countries, with an average yearly production of 161.8 million tons [1–3]. However, citrus greening disease, also called Huanglongbing (HLB), is becoming a global warning to citrus species sustainability, predicting a severe economic loss to the multi-billion dollar citriculture industry [4]. HLB is linked with intracellular

Candidatus Liberibacter species (CLs) transmitted mainly by Asian *Diaphorina citri* and African *Trioza erytreae* citrus psyllids [5,6]. The disease leads to symptoms of phloem disorder and root decay, changing the plant source-sink associations, rapid tree weakening, and affecting fruit production. Some signs of the disease include small-sized fruit, off-colored, irregular shape, bad-tasting fruit, and blotchy leaves (Fig. 1). The bacterium *Candidatus Liberibacter* (CLs) resides in the phloem of diseased plants [4,6,7].

The strains of CLs responsible for HLB disease are *asiaticus*, *africanus*,

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and *americanus* [4]. Since there is no cure for HLB-infected trees yet, the spread of the disease is mainly controlled by removing infected trees and eradicating insect vectors [4,6]. Therefore, identifying infected plants on time and accurately is crucial for managing disease. At the same time, symptoms-based identification of diseased plants has poor precision owing to the resemblance of symptoms with other nutrient deficiencies and citrus diseases [6]. Also, detecting infected plants by nucleic acid-based assay and disease symptoms is not successful because nucleic acid-based assays involve complex sample preparation and expensive chemicals and are time-consuming, making them unsuitable for disease management, specifically when numerous samples of a single tree have to be tested [4,7,8]. However, detecting infected plants by secreted protein-based biomarkers is an alternative as it is accurate, speedy, robust, and portable. These secreted proteins are systematically and evenly distributed throughout the infected plants, therefore excluding the requirement of multiple sampling from the same plant. Also, the systematic distribution of secreted proteins makes this technique reliable because it reduces the false negative results [9].

2. Disease symptoms

2.1. Leaf symptoms

The leaves of the HLB-diseased plants have a pattern of asymmetrical blotchy mottling and chlorosis, similar to the Fe deficiency symptom when leaves become stained yellow [10]. However, the symmetrical pattern can be found if trees suffer from nutrient deficiency and other disorders. Usually, these types of nutrient deficiency and blotchy mottles symptoms are observed in the plants affected by HLB. In contrast, these symptoms cannot be considered absolute HLB symptoms. The blotchy mottles can be present in mature leaves throughout the tree canopy. Symptoms similar to yellow and corky veins alone cannot be considered diagnostic of citrus greening. If these symptoms are present, then the whole tree canopy should also be checked for mottle symptoms such as the presence of bright yellow shoots, corky mid-veins, yellowing of under-rib, and interveinal chlorosis in leaves [10,11].

2.2. Fruit symptoms

After infection, the HLB-diseased plants deteriorate fast, resulting in the appearance of disease signs on the fruits, too. After healthy fruits reach ripeness, the stylus end becomes yellow/orange and the peduncular end green. However, in fruits with HLB disease, the stylus and the peduncular end color are green and yellow-orange, respectively [12]. The HLB-infected fruits can be ununiform in shape, smaller in size, and green in color depending on the different cultivars of fruits and their development stages. In addition, after an HLB infection, fruit falling begins in its early stage of life [13]. Even though several of these symptoms can be mistaken for nutrient deficiencies and disorders, the HLB infection symptoms are generally evident at first in one branch or a part of a tree and extend throughout the whole canopy. However, nutrient deficiencies and disorders are limited among trees and are typically more uniform.

2.3. Whole tree symptoms

The blotch mottle and yellow veins are observed in the yellow shoot and narrow and erect leaves. Also, various symptoms like a yellow look, stem dieback, slow growth, and fruit drops have been detected. First, the symptoms are only found in some parts of the canopy and spread afterward. It is also seen that the roots of CLS-infected trees are destroyed. Therefore, the root of the HLB-infected diseased plants is significantly affected, resulting in a loss of whole plants [14]. The root decay generally initiates long before the appearance of visible symptoms, resulting in a decline in the root-shoot ratio in the HLB-diseased plant with time [15]. Shahzad et al. [16] reported that the root, shoot, and leaves biomass are higher in healthy plants compared to the HLB-infected plants, and 40–50 % of lower root biomass was observed in the infected plants.

3. Differences between HLB and other nutrient deficiency problems

Since mineral deficiency symptoms are similar to citrus greening symptoms, it is crucial to identify them correctly because zinc, iron,

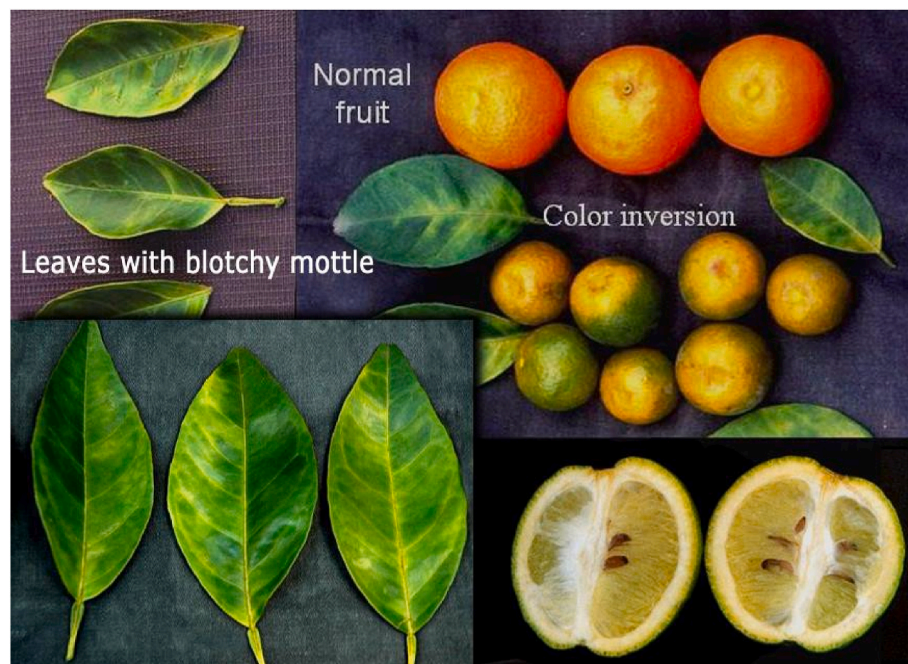


Fig. 1. HLB disease symptoms of rapid tree decrease, where fruit yield by the infected tree is considerably affected. Symptoms include small, poorly colored, lopsided, and off-tasting fruit, and mottled leaves (National Academies Press, 2010).

manganese, and calcium deficiencies also cause the yellowing of leaves. However, precise identification could isolate HLB-diseased plants from mineral-deficient plants. In HLB-diseased plants, a diffuse asymmetric mottle is produced in the leaves that show a yellow patch and fade green compared to green leaves in healthy plants. In mineral-deficient plants, each mineral deficiency exhibits different leaf patterns, but in all type of mineral deficiency, the patterns start from the mid-vein region of leaves [17]. In the zinc (Zn) deficiency case, an interveinal yellow mottle is found in each leaf. In the case of severe Zn deficiency, the leaves are small, narrow, and chlorotic. Manganese (Mn) deficiency symptoms are similar to zinc deficiency, but the leaf colors are a combination of light green and dark green. In the case of magnesium deficiency, an upturned "V" form of leaves is seen with green and yellow patches on both sides of the central vein of the leaves. The iron-deficient plants have leaves that are light green and pale yellow. In the case of calcium deficiency, fading of chlorophyll occurs along the margin of leaves, and small necrotic spots can be observed in the faded areas. Calcium deficiency can also generate small and thick leaves that initiate loss of vigor, foliage thinning, and reduced fruit production. Nitrogen deficiency can cause small size and yellowing of whole leaves [17]. Also, unlike nutrient deficiencies that typically cause symmetrical yellowing across the leaf, HLB often causes blotchy, asymmetrical yellowing. Additionally, leaves affected by HLB may show thickened, corky veins. Fruit from HLB-infected trees is often small, green, and misshapen with a bitter taste. These symptoms do not occur with nutrient deficiencies. While nutrient deficiencies can be corrected, HLB leads to progressive decline of the tree's health and eventual death. For precise and efficient identification, experts perform visual inspection regularly based on the aforementioned aspects before they sent leaf samples to a laboratory for molecular analysis.

4. HLB-disease-causing bacteria (CLs)

Candidatus Liberibacter species (CLs) are associated with HLB disease in citrus plants. Despite several efforts, pure cultures of CLs have not yet been created, and maximum data regarding the biology of CLs has been gathered from their genome sequences [18,19]. The genome analysis from six CLs strains provided information about hundreds of metabolisms and transport protein-related genes [20]. The grouping of genome analysis and RNA-Seq data from CLs-infected hosts and infected plants have disclosed particular metabolic characteristics controlled by the CLs gene in every Citrus host type. Through genome analysis and RNA-Seq data, target genes can be identified to be inhibited to block bacterial growth. From the above data, six CLs transcriptional regulator genes (RNA polymerase sigma factor-*RpoH*, vital for swimming global activators-*VisNR*, MarR family of transcriptional regulators-*LdtR*, auto-inducer 2-binding-*LsrB*, putative transcription factor-*PhrR*, and response regulator transcription factors-*CtrA*) have been identified to be crucial for CLs survival and replication [21]. Barnett et al. [21] also suggested that CLs types of parasites with small genome size may also depend on the related microbiota to obtain and make essential metabolites [22]. Therefore, the research on phloem-related microbiota may expose additional target genes or organisms to limit CLs. More than 150 known secretory pathway (Sec)-dependent effectors (SDEs) have been depicted for different species of CLs [23]. The role of some SDEs has been studied by assessing the gene expression levels in different CLs-infected samples or by bacterial and plant surrogate models [21,24]. However, changes in the host reactions and pathogenic methods of surrogate systems and CLs have not yet been confirmed in their natural hosts; hence, the results can not be completely reliable [22]. Only three SDEs have been studied via stable transformation of Citrus hosts. SDE1 and SDE15 were observed to improve CLs infection once overexpressed in transgenic grapefruit. The SDE1 reduced the activity of papain-like cysteine protease (PLCP) and plant defense. Meanwhile, SDE15 acted on accelerated cell death 2 (ACD2), a susceptibility-related gene for CLs [25,26]. Like SDE1 and SDE15, the third SDEs (P235) overexpression induced HLB-like

symptoms in transgenic citrus plants [27]. However, these studies are laborious in getting transgenic lines and accomplishing regular CLs inoculations, making scaleup impossible. Recently, a rapid antimicrobial testing method has been developed involving *Rhizobium rhizogenes*-mediated induction of hairy root development by healthy or CLs-infected citrus explants [28]. The roots are then transferred to culture plates, and by analyzing the variations in CLs multiplication in healthy or CLs-infected citrus explants, 6 antimicrobial peptides and 8 growth inhibitors were identified. This assay is much faster (it needs 4–6 weeks to finish) than the conventional method. Furthermore, as *R. rhizogenes*-mediated genetic transformation in citrus plants can be performed easily, this hairy root methodology will also permit the execution of genetic tests to identify the possible host target genes responsible for CLs resistance [28].

5. Transmission vectors of HLB-associated CLs

The entomological and biological aspects of CLs vector Asian citrus psyllid (*D. citri*) are well studied [29]. The studies on *D. citri* over the last decade revealed that this vector can distinguish between dissimilar host blends by the signal of specific volatile compounds (VOCs) of the host plants. The *D. citri*-repellent VOCs were identified in guava using electrophysiological and behavioral assays by Rouseff et al. [30]. Recently, 5 Chinese plants have been found to secrete more potent repellent VOCs for *D. citri* [31]. The VOCs for *D. citri* repellent could be utilized to modify Citrus plants at the molecular level to make them repellent to *D. citri*. In another way, by modulating the production of specific VOCs liable for *D. citri* attraction, less affinity to *D. citri* citrus plant can be obtained. Psyllid insects also have an affinity to bright yellow and green colors, which may support the presence of *D. citri* in non-host trees [32]. In addition, it is demonstrated that *D. citri* can not access the phloem of non-host plants even though they reach the xylem of non-host plants [33]. This might be due to the compatibility response between HLB-carrying psyllids and host plants of other phloem feeders of the host plants. Thus, the interaction between the host defense system and the effector molecules discharged in the insect saliva plays a vital role in plant-insect relations. The genomic and proteomic data from *D. citri* saliva and salivary glands are being used to detect potential effectors for plant-insect relations [34]. Recently, RNA interference (RNAi) technology has been employed to target the critical genes of *D. citri* to control psyllid populations [35,36]. By targeting multiple genes of the psyllid, insect mortality could be increased, and the spread of CLs could be decreased. Another biological control is using *Bacillus thuringiensis*, which can induce high *D. citri* mortality. *Bacillus thuringiensis* produces different Cry proteins responsible for insect toxicity, and it has been found that Cry1Ab is most effective against *D. citri* [37]. Therefore, the transgenic overexpression of Cry1Ab toxins targeted to the phloem of the host plant could generate psyllid-killing citrus plants.

6. Interaction of host plants with HLB-associated CLs and transmission vectors

The resistance of citrus trees to HLB might be due to their resistance against CLs or psyllid insects or against both. CLs and psyllid insects have a broad host range that covers the maximum number of species of Aurantioideae. Aurantioideae species include all genotypes of Citrus, including Poncirus and Fortunella and their hybrids with Citrus. Poncirus and Fortunella are relevant species because these are extensively used as rootstock for citriculture and ornamental functions. Even though citrus cultivars demonstrated distinctive sensitivity to CLs infection, all cultivars allow bacterial replication and affect fruit production. For instance, the level of damage caused by CLs to Eureka lemons and Tahiti limes is lower than that caused by orange and mandarin cultivars [38]. The lack of resistance against CLs in citrus plants might also be due to their clonal propagation. However, some Aurantioideae species have been observed to be resistant to CLs [39]. In addition, some Oceanian

Citrus relatives from the Eremocitrus and Microcitrus exhibited complete resistance to CLs infection, as they inhibit bacterial multiplication. It was supposed that the resistance of Eremocitrus and Microcitrus spp to CLs might be due to the deactivation of susceptibility genes that happened after the Asian and Oceanian species diverged [39]. Some other species of Citrinae subtribe (sexually incompatible with Citrus) also exhibited partial resistance to CLs. Aurantioideae species are resistant to CLs but are highly susceptible to *D. citri*. This species can be trap-and-kill *D. citri* [40]. However, psyllids demonstrate differences in their inclination for distinct Citrus species; none of these plant genotypes have been designated fully resistant to psyllids [41].

7. Management practices for HLB control

The management practices to control HLB and their advantages and disadvantages are summarized in Table 1.

7.1. Control of HLB by antibiotics

The uses of antibiotics to manage plant diseases have been a topic of debate among researchers for years. The extensive use of antibiotics is debatable owing to public worries concerning their probable side effects on human health and the emergence of antibiotic-resistant microorganisms [42,43]. Although certain antibiotics have been observed to be effective in managing plant diseases, they have had to be stopped partly due to their high costs in the field. In the 1970s, the application of tetracycline against plant disease was discontinued due to high price and phytotoxicity [44]. Streptomycin and oxytetracycline were also suspended after more than 50 years of use to bacterial plant disease in the United States because of their high price and the emergence of antibiotic resistance bacteria [45,46]. Various researchers have reported using antibiotics through foliar sprays, soil drenches, and trunk injections to control HLB disease [47]. Among these methods, using antibiotics through trunk injections was more effective than foliar sprays and soil drenches [48]. Meanwhile, the high price of the trunk injection method of antibiotic application was the limiting factor for large-scale use. Shokrollah et al. [49] found that injecting antibiotics with gibberellic acid 3 in infected citrus plants increased the fruit yield by 231 % compared to untreated plants. Also, Hu et al. [50] found a 15 % increase in the fruit yield when infected citrus plants were treated with oxytetracycline using trunk injection with variable numbers of injection ports. Hu et al. [50] reported a significant increase in fruit yield and fruit

quality in sweet orange trees when 8 plant activators were applied with 3 antibiotics (oxytetracycline, penicillin G, and streptomycin) through trunk injection. Although the use of penicillin, oxytetracycline, and streptomycin sulfate increased growers' profits by \$339–\$664 per acre, these antibiotics persist in fruits and pose a risk for human intake. Particularly, Penicillin was prohibited for its use in citrus orchards because of the potential risk to human health and bacterial-resistant development [51,52].

On the other hand, using a foliar spray of plant regulators results in a loss to growers because of the high cost of treatments. In 2016, streptomycin and oxytetracycline complex treatment was approved to treat HLB in the form of a foliar spray regardless of the absence of evidence of such methods being successful in the field [45,50]. It has been observed that foliar spray did not positively affect fruit yield. New studies have recommended that antimicrobial peptides are favorable in managing HLB disease, but the high price of antimicrobial peptides has delayed their use in the citrus field [53,54].

7.2. Application of insecticides

Broad-spectrum insecticides have been used successfully to control adult citrus psyllids, specifically during the plant dormancy period [55]. Systemic insecticides such as neonicotinoid, thiamethoxam, imidacloprid, clothianidin, and cyantraniliprole are allowed on only mature trees in citrus production. These insecticides are prohibited on young trees (Boina and Bloomquist, 2015; Qureshi et al., 2014) [56,57]. Stansly et al. [58] reported an increase in yield of 4 kg per Valencia orange tree (\$279 per acre) in after applying seven insecticides to control the ACP. Stansly et al. [58] also found an increase in the yield by 6 kg per tree after insecticide treatment. The cost of insecticide applications was around \$99–\$278 per acre. Monzo and Stansly [59] demonstrated that insecticide application generates higher income (~\$985 per acre). Tansey et al. [60] also found a 20 % higher yield (80 kg/tree) in insecticide-treated trees (13 different insecticide treatments) compared to 67 kg/tree for untreated trees. Monzo and Stansly [61] used 30 insecticides and found significant improvement in yield, but the cost of insecticide treatment was much higher (\$607/acre). The studies outside North America also exhibited similar results regarding the cost of insecticide treatment (Belasque et al., 2010; Lopez and Durborow, 2015) [62,63]. Although the insecticide treatment to control psyllids has shown a higher yield of citrus fruit, the intensive use of insecticide can induce resistance in psyllids, resulting in secondary pest outbreaks [60]. For example, Chen et al. [64] found moderate resistance against thiamethoxam in psyllid populations. The economic threshold is a critical Integrated Pest Management practice for applying insecticides to control the possibility of resistance occurrence and outbreak of secondary pests [61].

7.3. Improved foliar nutritional programs

Many citrus fruit growers have employed improved foliar nutritional programs (FNP) in their orchards [38,58]. Some research revealed that FNP can sustain the yield of HLB-infected trees by alleviating the nutritional deficiencies triggered by HLB [65,66]. Nutrients like nitrogen, phosphorus, potassium, manganese, zinc, boron, and magnesium are extensively used for the growth and higher yield of CLs-infected citrus trees [67]. The macronutrients (nitrogen, phosphorus, and potassium salts) and micronutrients (zinc, boron, and magnesium) allow leaves to obtain essential elements that might be reduced by root dysfunction. Stansly et al. [58] reported a higher fruit yield in HLB-affected Valencia orange trees after FNP. However, the use of FNP has also been criticized due to its cost, which is higher than the increased yield of the fruits [60]. Morgan et al. [67] report that the average yield of the 3-year application of MnSO₄ was marginally higher than unsprayed trees. Rouse et al. [66] found improved early yield of citrus fruits by spraying a mixture of micronutrients, macronutrients, potassium nitrate, and urea. Stansly et al. [58] applied FNP to alleviate the

Table 1
Management practices to control HLB in citrus.

Method	Advantage	Disadvantage
Antibiotics	Very effective in controlling HLB	High price and emergence of antibiotic resistance microbes
Antimicrobial peptide	Effective in managing HLB	High price
Insecticide	Control adult citrus psyllids during the plant dormancy period and higher yield	Induce resistance in psyllids and cause secondary pest outbreaks
Foliar nutritional programs	Sustain the yield of HLB-infected trees by alleviating the nutritional deficiencies triggered by HLB	Higher cost than increased yield
Thermotherapy	Very effective in controlling CLs and HLB	Cause harsh damage to plant tissues and difficult to implement in field condition
Biological control	Effective in controlling HLB	Difficult to estimate cost effectiveness due to varying price of biological agent in different countries
Tree removal or pruning	Effective in managing psyllids populations	Costly and not easy to execute
Transgenic plant	Plants become less susceptible to CLs	Expensive and non-acceptance of transgenic by consumers

devastating effects of the HLB disease and found the net income was 4% per acre. They reported a change in fruit yield from -24 % to 23 % with an increased cost of \$729 per acre when a mixture of micronutrients and macronutrients was used. On the other hand, Ozores-Hampton et al. [65] reported that the FNP did not show bactericidal effects against CLs and only increased the yield by 7 %–9 %, and the cost associated with the FNP package was \$343/acre. Most studies demonstrated that FNPs are not economically cost-effective.

7.4. Thermotherapy

Thermotherapy treatment has been recommended to control CLs infection [68]. Hoffman et al. [69] reported around 40-fold decrease in CLs titer in grapefruit when treated at 40 °C for 24 h or 42 °C for 19 h for 2 successive days. Zhang et al. [70] reported a significant reduction in CLs titer when HLB-infected Ray Ruby seedlings were treated at 45 °C for 8 h per day for 1 week. Similarly, Fan et al. [71] reported a 30 % and 55 % reduction of CLs in leaves when seedlings were treated at 45 °C and 48 °C for 4 h for 1 day/week for 3 consecutive weeks, respectively; the thermotherapy caused harsh damage to plant tissues. Moreover, the primary challenge associated with thermotherapy is that its application in the field is complex. Also, the optimum temperature and duration to kill CLs are still uncertain [72]. If thermotherapy is applied in the field conditions, the high-temperature treatment may injure the leaves; however, the heat treatment may not affect the root system and thick portion of the bark on the trunk, thus providing inoculum to reinfect the trees [71] (Fan et al., 2016). Therefore, thermotherapy can be efficient if the plants are adequately treated above and below the grounds (treatment of root and stems) [45]. Yang et al. [68] reported that the combined therapy of antimicrobial compounds and thermotherapy increased the delivery efficacy of antimicrobial compounds into the citrus phloem. Al-Jumaili and Ehsani [73] found that the juice quality of fruits changed significantly when the air temperature was increased to 55–65 °C for about 1–2 min. While the juice acidity was not affected by heat treatments, the brix and acid ratio increased considerably at an elevated temperature. Ehsani et al. [74] covered Valencia orange tree orchards with translucent plastic for 5 h in midsummer to heat by the sunlight to 45 °C and found 30 % higher yields than controlled trees. In one study, Ehsani and Pertiwi [72] reported a higher yield in trees treated at 55 °C for 1 h than in non-treated trees. However, in another study, Ehsani et al. [75] recommended that heat treatments had antagonistic effects on the yield in the first year due to heat-induced fruit drop. Dewdney et al. [76] also reported that thermotherapy does not improve yield enough to compensate for the damage caused to the canopy. Li et al. [77] documented the cost of thermotherapy for controlling HLB which includes the fixed cost of the thermotherapy system included mobile platform (\$15,000), steam generator (\$20,000), water storage (\$285), water pump (\$1000), water filter (\$60), water softener (\$2300), electricity generator (\$600), tree enclosure (\$15,000), and electronics (\$600), for a total of \$54,845. The variable cost for operation primarily consisted of labor and fuel, which is \$2.40 per tree on average. The total cost was \$6.50–\$7.50 per tree if renting a machine. Therefore, it is concluded that the heat treatment adversely affects the fruit yield.

7.5. Biological control

CLs cause the HLB disease in the citrus plant, and psyllid is the vector for transmission of CLs. Therefore, the control of vectors could be essential for managing HLB. Applying biological control in combination with organic improvements, resistant varieties, or pathogen-free seedlings could be useful [78]. *Tamarixia radiata* is a wasp used as a biocontrol agent against psyllid. *T. radiata* was introduced in and reduced the psyllid population by 4 %–70 % [4]. The beetle called *Hippodamia convergens* Guerin-Meneville has been released in North America to control psyllids, but it is not commonly used as a biocontrol agent in citrus groves yet [57]. Alvarez et al. (2016) [78] estimated the

cost of the biological control agents in Brazil and found the price to be \$9.71/acre, which is more than double that in Florida. To date, no studies have shown the effect of biological control on yield; therefore, its cost-effectiveness is not calculated in this study.

7.6. Other treatments

Despite constant attempts by citrus growers to increase their fruit production by controlling HLB, the CLs have spread to infect most citrus orchards [66]. Other proposed treatments for managing HLB infection include tree removal and pruning, interplanting crops, and transgenic biotechnology. According to some researchers, tree removal and pruning exhibited efficient treatment for managing psyllids populations. However, this is not a cost-effective approach and isn't easy to execute as it needs synchronized efforts from neighboring citrus growers [4]. Muraro [79] estimated about \$450/acre as an increased grove maintenance cost for removing infected trees. Similarly, Roka et al. [80] reported increased grove maintenance costs of \$200 to \$600 per acre. In addition, the study by Ozores-Hampton et al. [65] reported a higher yield in pruned trees than nonpruned trees, but the cost of pruning was \$160 per acre. Other researchers have suggested that interplanting guava plants with citrus plants can reduce the psyllids infestation in citrus [81]. They reported a significant decline in the Asian citrus psyllid population when citrus trees were interplanted with pink guava; however, no reduction was observed when citrus trees were interplanted with white guava. Moreover, interplanted citrus plants with either white or pink guava did not stop the introduction and spread of HLB. The resistance of *Eremocitrus* and *Microcitrus* against CLs and the harmful effects of specific *Poncirus* cultivars on psyllid development have encouraged breeding research expected to obtain HLB-resistant varieties. Though such breeding programs take several years to complete, they could be sped up if segregation progenies permit the detection of undiscovered molecular markers related to HLB resistance [82]. Otherwise, the genes associated with HLB resistance or sensitivity could be cloned to relevant commercial cultivars faster via transgenic approaches than conventional breeding. Presently, only a few transgenic approaches against HLB have been tested. Citrus cultivars producing antimicrobial peptides active against other pathogenic bacteria did not show strong CLs resistance [83]. The development of a citrus variety that overexpresses an NPR1 gene of *Arabidopsis thaliana* can regulate systemic acquired resistance for managing HLB. NPR-1 gene is linked to systemic acquired resistance in grapefruits and oranges, and it is found to be resistant against CLs to some extent [84]. In addition, transgenic plants expressing chimeric constructs of NPR-1, thionin, and antimicrobial peptides exhibited resistance against CLs and did not show leaf mottling [85]. Recently, Alquezar et al. [5] generated engineered orange plants resistant to *D. citri*. However, the effects of these approaches on the occurrence and spread of HLB in the field have yet to be reported. Although gene-editing tools, such as the CRISPRCas9, are being used to create citrus cultivars that are less susceptible to CLs or to express genes that can prevent vector transmission, transgenic biotechnology is costly [84]. It is also imperative to believe that implementing bioengineering on citrus will possibly face present challenges in the market of non-acceptance of transgenics by some consumers.

The cost of insecticide sprays ranges between \$100 and \$500 per acre, making it the cheapest treatment among all the major treatments analyzed. The costs of using antibiotics and nutritionals reported in the literature were similar and fluctuated between \$350 and \$660 per acre. With a cost of \$850–\$1000 per acre, thermotherapy represents the most expensive treatment proposed for coping with HLB.

8. Conclusions

The global expansion of Huanglongbing (HLB), or Citrus greening disease, has catalyzed a race among researchers to address its profound threat to the citriculture industry urgently. HLB, driven by *Candidatus*

Liberibacter species (CLs) transmitted by psyllid insects, severely cripples citrus plants, leading to compromised yields and plant mortality. Despite ongoing efforts, traditional control methods, including insecticides and antimicrobials, have shown limited effectiveness. Existing diagnostic techniques based on nucleic acid assays and symptom observation present challenges that hinder accurate and cost-effective disease detection. In response to these challenges, this review underscores the potential of secreted protein-based biomarkers as a breakthrough solution for rapid and precise HLB detection. These biomarkers could significantly enhance disease management strategies by offering a reliable and efficient means of disease identification. The review also emphasizes the importance of distinguishing HLB symptoms from nutrient deficiencies, highlighting the critical role of accurate identification in effective mitigation. The paper further addresses the complex interactions between CLs and host plants, acknowledging the obstacles posed by the elusive nature of these pathogens. While challenges persist in culturing CLs in axenic conditions, genome sequencing has illuminated their metabolic characteristics and potential virulence factors. Efforts to identify host target genes for resistance are discussed, along with introducing an innovative Citrus hairy roots-based high-throughput antimicrobial testing approach. Potential prospects for long-term solutions suggest transgenic citrus that overexpresses existing genes to regulate the acquired defense mechanism. In summary, the urgent need to combat HLB demands revolutionary solutions. By shedding light on the potential of secreted protein-based biomarkers and innovative testing methodologies, this paper serves as a rallying call to transform HLB management strategies. By embracing these advancements, the global citriculture industry can aspire to fortify its production and export.

CRediT authorship contribution statement

Alya Limayem: Writing – review & editing, Funding acquisition, Conceptualization. **Elizabeth M. Martin:** Writing – review & editing. **Shiv Shankar:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

All authors state that there is no conflict of interest.

Data availability

No data was used for the research described in the article.

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References

- [1] Food and agriculture organization statistics, retrieved from, <http://www.fao.org/3/a-i8092e.pdf>, 2016.
- [2] G. Zhong, E. Nicolosi, Citrus origin, diffusion, and economic importance, in: A. Gentile, S. La Malfa, Z. Deng (Eds.), *The Citrus Genome*. Compendium of Plant Genomes, Springer, Cham, 2020, pp. 5–21, https://doi.org/10.1007/978-3-030-15308-3_2.
- [3] M.P. Gonzatto, J.S. Santos, Introductory chapter: world citrus production and research, in: P.G. Mateus, S.S. Julia (Eds.), *Citrus Research-Horticultural and Human Health Aspects*, IntechOpen, 2023, <https://doi.org/10.5772/intechopen.110519>.
- [4] T.R. Gottwald, Current epidemiological understanding of citrus huanglongbing, *Annu. Rev. Phytopathol.* 48 (2010) 119–139, <https://doi.org/10.1146/annurev-phyto-073009-114418>.
- [5] B. Alquezar, L. Carmona, S. Bennici, L. Pena, Engineering of citrus to obtain huanglongbing resistance, *Curr. Opin. Biotechnol.* 70 (2021) 196–203, <https://doi.org/10.1016/j.copbio.2021.06.003>.
- [6] J.M. Bové, Huanglongbing: a destructive, newly-emerging, century-old disease of citrus, *J. Plant Pathol.* 88 (2006) 7–37.
- [7] D.C. Teixeira, C. Saillard, C. Couture, E.C. Martins, N.A. Wulff, S. Eveillard-Jagoueix, Distribution and quantification of *Candidatus Liberibacter americanus*, agent of huanglongbing disease of citrus in São Paulo State, Brasil, in leaves of an affected sweet orange tree as determined by PCR, *Mol. Cell. Probes* 22 (2008) 139–150, <https://doi.org/10.1016/j.mcp.2007.12.006>.
- [8] M.S. Irey, T. Gast, T.R. Gottwald, Comparison of visual assessment and polymerase chain reaction assay testing to estimate the incidence of the Huanglongbing pathogen in commercial Florida citrus, *Proc. Fla. State Hortic. Soc.* 119 (2006) 89–93.
- [9] D. Pagliaccia, J. Shi, Z. Pang, E. Hawara, K. Clark, S.P. Thapa, A pathogen secreted protein as a detection marker for citrus huanglongbing, *Front. Microbiol.* 8 (2017) 2041, <https://doi.org/10.3389/fmicb.2017.02041>.
- [10] H. Inoue, S. Yamashita-Muraki, K. Fujiwara, K. Honda, H. Ono, T. Nonaka, Y. Kato, T. Matsuyama, S. Sugano, M. Suzuki, Y. Masaoka, Fe²⁺ ions alleviate the symptom of citrus greening disease, *Int. J. Mol. Sci.* 21 (2020) 4033, <https://doi.org/10.3390/ijms21114033>.
- [11] E.D. Ammar, D.G. Hall, R.G. Shatters Jr, Stylet morphometrics and citrus leaf vein structure in relation to feeding behavior of the Asian citrus psyllid *Diaphorina citri*, vector of citrus huanglongbing bacterium, *PLoS One* 8 (2013) e59914, <https://doi.org/10.1371/journal.pone.0059914>.
- [12] J.M. Bove, Huanglongbing or yellow shoot, a disease of Gondwanan origin: will it destroy citrus worldwide? *Phytoparasitica* 42 (2014) 579–583, <https://doi.org/10.1007/s12600-014-0415-4>.
- [13] B.M. Dala-Paula, A. Plotto, J. Bai, J.A. Manthey, E.A. Baldwin, R.S. Ferrarezi, M.B. A. Gloria, Effect of huanglongbing or greening disease on orange juice quality, a review, *Front. Plant Sci.* 9 (2019) 1976, <https://doi.org/10.3389/fpls.2018.01976>.
- [14] J.H. Graham, E.G. Johnson, T.R. Gottwald, M.S. Irey, Presymptomatic fibrous root decline in citrus trees caused by huanglongbing and potential interaction with *Phytophthora* spp, *Plant Dis.* 97 (2013) 1195–1199, <https://doi.org/10.1094/PDIS-01-13-0024-RE>.
- [15] E.G. Johnson, J. Wu, D.B. Bright, J.H. Graham, Association of ‘*Candidatus Liberibacter asiaticus*’ root infection, but not phloem plugging with root loss on huanglongbing-affected trees prior to appearance of foliar symptoms, *Plant Pathol.* 63 (2014) 290–298, <https://doi.org/10.1111/ppa.12109>.
- [16] F. Shahzad, C. Chun, A. Schumann, T. Vashisth, Nutrient uptake in huanglongbing-affected sweet orange: transcriptomic and physiological analysis, *J. Am. Soc. Hortic. Sci.* 145 (2020) 349–362, <https://doi.org/10.21273/JASHS04929-20>.
- [17] M.M.H. Tipu, M.M. Masud, R. Jahan, A. Baroi, A.K.M.A. Hoque, Identification of citrus greening based on visual symptoms: a grower’s diagnostic toolkit, *Heliyon* 7 (2021) e08387, <https://doi.org/10.1016/j.heliyon.2021.e08387>.
- [18] J.F. Coyle, G.L. Lorca, C.F. Gonzalez, Understanding the physiology of *Liberibacter asiaticus*: an overview of the demonstrated molecular mechanisms, *J. Mol. Microbiol. Biotechnol.* 28 (2018) 116–127, <https://doi.org/10.1159/000492386>.
- [19] M.V. Merfa, P. Edel, Progress and obstacles in culturing ‘*Candidatus Liberibacter asiaticus*’, the bacterium associated with Huanglongbing, *Phytopathology* 109 (2019) 1092–1101, <https://doi.org/10.1094/PHYTO-02-19-0051-RVW>.
- [20] C. Zuniga, B. Peacock, B. Liang, G. Mccollum, S.C. Irigoyen, D. Teccampas, C. Marotz, N. Weng, A. Zepeda, G. Vidalakis, Linking metabolic phenotypes to pathogenic traits among ‘*Candidatus Liberibacter asiaticus*’ and its hosts, *NPJ Syst. Biol. Appl.* 6 (2020) 24, <https://doi.org/10.1038/s41540-020-00142-w>.
- [21] M.J. Barnett, D.E. Solow-Cordero, S.R. Long, A high-throughput system to identify inhibitors of *Candidatus Liberibacter asiaticus* transcription regulators, *Proc. Natl. Acad. Sci. USA* 116 (2019) 18009–18014, <https://doi.org/10.1073/pnas.1905149116>.
- [22] Y. Jiang, C. Zhang, R. Chen, S. Yang, Challenging battles of plants with phloem-feeding insects and prokaryotic pathogens, *Proc. Natl. Acad. Sci. USA* 116 (2019) 23390–23397, <https://doi.org/10.1073/pnas.1915396116>.
- [23] Q. Shi, M. Pitino, S. Zhang, J. Krystel, L.M. Cano, G.B.R. Shatters Robert, D.G. Hall, E. Stover, Temporal and spatial detection of *Candidatus Liberibacter asiaticus* putative effector transcripts during interaction with -resistant citrus hosts, *BMC Plant Biol.* 19 (2019) 122–133, <https://doi.org/10.1186/s12870-019-1703-4>.
- [24] C.Y. Huang, D.D. Niu, G. Kund, M. Jones, U. Albrecht, L. Nguyen, C. Bui, C. Ramadugu, K.D. Bowman, J. Trumble, Identification of citrus immune regulators involved in defence against Huanglongbing using a new functional screening system, *Plant Biotechnol. J.* 19 (2020) 757–766, <https://doi.org/10.1111/pbi.13502>.
- [25] K.J. Clark, Z. Pang, J. Trinh, N. Wang, W. Ma, Sec-delivered effector 1 (SDE1) of ‘*Candidatus Liberibacter asiaticus*’ promotes citrus Huanglongbing, *Mol. Plant Microbe Interact.* 33 (2020) 1394–1404, <https://doi.org/10.1094/MPMI-05-20-0123-R>.
- [26] Z. Pang, L. Zhang, G. Coaker, W. Ma, S.Y. He, N. Wang, Citrus CsACD2 is a target of *Candidatus Liberibacter asiaticus* in huanglongbing disease, *Plant Physiol.* 184 (2020) 792–805, <https://doi.org/10.1104/pp.20.00348>.
- [27] G. Hao, D. Ammar, Y. Duan, E. Stover, Transgenic citrus plants expressing a ‘*Candidatus Liberibacter asiaticus*’ prophage protein LasP 235 display Huanglongbing-like symptoms, *Agri. Gene.* 12 (2019) 1–10, <https://doi.org/10.1016/j.aggene.2019.100085>.
- [28] S. Irigoyen, M. Ramasamy, S. Pant, P. Niraula, R. Bedre, M. Gurung, D. Rossi, C. Laughlin, Z. Gorman, D. Achor, Plant hairy roots enable high throughput identification of antimicrobials against *Candidatus Liberibacter* spp, *Nat. Commun.* 11 (2020) 5802, <https://doi.org/10.1038/s41467-020-19631-x>.

- [29] B. Alquezar, H.X.L. Volpe, R.F. Magnani, M.P. Miranda, M.A. Santos, V. Marques-Vieira, M.R. Almeida, N.A. Wulff, H.M. Ting, M. Vries De, Engineered orange ectopically expressing the Arabidopsis b-caryophyllene synthase is not-attractive to *Diaphorina citri*, the vector of the bacterial pathogen associated to Huanglongbing, *Front. Plant Sci.* 12 (2021) 200–214, <https://doi.org/10.3389/fpls.2021.641457>.
- [30] R.L. Rouseff, E.O. Onagbola, J.M. Smoot, L.L. Stelinski, Sulfur volatiles in guava (*Psidium guajava* L.) leaves: possible defense mechanism, *J. Agric. Food Chem.* 56 (2008) 8905–8910, <https://doi.org/10.1021/jf801735v>.
- [31] Z. Yan, Q. Zhang, N. Zhang, W. Li, C. Chang, Y. Xiang, C. Xia, T. Jiang, W. He, J. Luo, Repellency of forty-one aromatic plant species to the Asian citrus psyllid, vector of the bacterium associated with Huanglongbing, *Ecol. Evol.* 10 (2020) 12940–12948, <https://doi.org/10.1002/ece3.6876>.
- [32] N. Johnston, P.A. Stansly, L. Stelinski, Secondary hosts of the Asian citrus psyllid, *Diaphorina citri* Kuwayama: survivor and preference, *J. Appl. Entomol.* 143 (2019) 921–928, <https://doi.org/10.1111/jen.12673>.
- [33] J. George, R. Kaniserry, E.D. Ammar, I. Cabral, L.T. Markle, J.M. Patt, L. Stelinski, Feeding behavior of Asian citrus psyllid [*Diaphorina citri* (Hemiptera: Liviidae)] nymphs and adults on common weeds occurring in cultivated citrus using electrical penetration graph recordings, *Insects* 11 (2020) 48, <https://doi.org/10.3390/insects11010048>.
- [34] X.Q. Liu, H.B. Jiang, T.Y. Liu, L. Yang, J.Y. Fan, Y. Xiong, T.X. Jing, B.H. Lou, W. Dou, J.J. Wang, A transcriptomic and proteomic analysis of the *Diaphorina citri* salivary glands reveals genes responding to *Candidatus Liberibacter asiaticus*, *Front. Physiol.* 11 (2020) 1–11, <https://doi.org/10.3389/fphys.2020.582505>.
- [35] W.B. Hunter, X.H. Sinisterra-Hunter, Emerging RNA suppression technologies to protect citrus trees from citrus greening disease bacteria, *Adv. Insect Physiol.* 55 (2018) 163–199, <https://doi.org/10.1016/b.s.aip.2018.08.001>.
- [36] S. Hajeri, N. Killiny, C. El-Mohhtar, W.O. Dawson, S. Gowda, Citrus tristeza virus-based RNAi in citrus plants induces gene silencing in *Diaphorina citri*, a phloem-sap sucking insect vector of citrus greening disease (Huanglongbing), *J. Biotechnol.* 176 (2014) 42–49, <https://doi.org/10.1016/j.jbiotec.2014.02.010>.
- [37] S. Dorta, O. de, J. Balbinotte, R. Monnerat, J.R.S. Lopes, T. da Cunha, O.Z. Zanardi, M.P. de Miranda, M.A. Machado, J. de Freitas-Astua, Selection of *Bacillus thuringiensis* strains in citrus and their pathogenicity to *Diaphorina citri* (Hemiptera: Liviidae) nymphs, *Insect Sci.* 27 (2020) 519–530, <https://doi.org/10.1111/1744-7917.12654>.
- [38] T. Gottwald, G. Poole, T. McCollum, D. Hall, J. Hartung, J. Bai, W. Luo, D. Posny, Y.P. Duan, E. Taylor, Canine olfactory detection of a vectored phyto-bacterial pathogen, *Liberibacter asiaticus*, and integration with disease control, *Proc. Natl. Acad. Sci. USA* 117 (2020) 3492–3501, <https://doi.org/10.1073/pnas.1914296117>.
- [39] M.N. Alves, S.A. Lopes, L.L. Raiol-Ju'nior, N.A. Wulff, E.A. Girardi, P. Ollitrault, L. Pena, Resistance to '*Candidatus Liberibacter asiaticus*' the Huanglongbing associated bacterium, in sexually and/or graft-compatible Citrus relatives, *Front. Plant Sci.* 11 (2020) 2166, <https://doi.org/10.3389/fpls.2020.617664>.
- [40] J.C. Cifuentes-Arenas, G.A. Charles Beattie, L. Pena, S.A. Lopes, *Murraya paniculata* and *Swinglea glutinosa* as short-term transient hosts of '*Candidatus Liberibacter asiaticus*' and implications for the spread of Huanglongbing, *Phytopathology* 109 (2019) 2064–2073, <https://doi.org/10.1094/PHYTO-06-19-0216-R>.
- [41] D.G. Hall, C. Ramadugu, M.G. Hentz, F.G. Gmitter, E. Stover, Survey of *Poncirus trifoliata* hybrids for resistance to colonization by Asian citrus psyllid, *Fla. Entomol.* 102 (2019) 635–637, <https://doi.org/10.1653/024.102.0339>.
- [42] S.G. Acimovic, Q. Zeng, G.C. McGhee, G.W. Sundin, J.C. Wise, Control of fire blight (*Erwinia amylovora*) on apple trees with trunk-injected plant resistance inducers and antibiotics and assessment of induction of pathogenesis-related protein genes, *Front. Plant Sci.* 6 (2015) 1–10, <https://doi.org/10.3389/fpls.2015.00016>.
- [43] V.O. Stockwell, B. Duffy, Use of antibiotics in plant agriculture, *Rev. Sci. Tech. Int. Epiz.* 31 (2012) 199–210, <https://doi.org/10.20506/rst.31.1.2104>.
- [44] F. Aldeek, D. Canzani, M. Standland, M.R. Crosswhite, W. Hammack, G. Gerard, J. M. Cook, Identification of penicillin G metabolites under various environmental conditions using UHPLC-MS/MS, *J. Agric. Food Chem.* 64 (2016) 6100–6107, <https://doi.org/10.1021/acs.jafc.5b06150>.
- [45] R.A. Blaustein, G.L. Lorca, M. Teplitski, Challenges for managing *Candidatus Liberibacter* spp. (huanglongbing disease pathogen): current control measures and future directions, *Phytopathology* 108 (2018) 424–435, <https://doi.org/10.1094/PHYTO-07-17-0260-RVW>.
- [46] G.W. Sundin, N. Wang, Antibiotic resistance in plant-pathogenic bacteria, *Annu. Rev. Phytopathol.* 56 (2018) 161–180, <https://doi.org/10.1146/annurev-phyto-080417-045946>.
- [47] T. Puttamuk, S. Zhang, Y. Duan, A. Jantasorn, N. Thaveechai, Effect of chemical treatments on '*Candidatus Liberibacter asiaticus*' infected pomelo (*Citrus maxima*), *Crop Protect.* 65 (2014) 114–121, <https://doi.org/10.1016/j.cropro.2014.07.018>.
- [48] M. Zhang, C.A. Powell, L. Zhou, Z. He, E. Stover, Y. Duan, Chemical compounds effective against the citrus huanglongbing bacterium '*Candidatus Liberibacter asiaticus*' in planta, *Phytopathology* 101 (2011) 1097–1103, <https://doi.org/10.1094/PHYTO-09-10-0262>.
- [49] H.S.H. Shokrollah, T.L. Abdullah, K. Sijam, A.A. Siti Nor, Identification of physical and biochemical characteristic of Mandarin (*Citrus reticulata*) fruit infected by huanglongbing (HLB), *Aust. J. Crop. Sci.* 5 (2011) 181–186, <https://www.cabidigitallibrary.org/doi/full/10.5555/20113087997>.
- [50] J. Hu, J. Jiang, N. Wang, Control of citrus huanglongbing via trunk injection of plant defense activators and antibiotics, *Phytopathology* 108 (2017) 186–195, <https://doi.org/10.1094/PHYTO-05-17-0175-R>.
- [51] C. Yang, C.A. Powell, Y. Duan, R. Shatters, Antimicrobial nanoemulsion formulation with improved penetration of foliar spray through citrus leaf cuticles to control citrus huanglongbing, *PLoS One* 10 (2015) e0133826.
- [52] M. Zhang, Y. Guo, C.A. Powell, M.S. Doud, C. Yang, Y. Duan, Effective antibiotics against '*Candidatus Liberibacter asiaticus*' in HLB affected citrus plants identified via the graft-based evaluation, *PLoS One* 9 (2014) e111032.
- [53] M.A. Guerra-Lupian, R. Ruiz-Medrano, J.A. Ramírez-Pool, F.-A. Ramírez-Ortega, J.-A. Lopez-Buenfil, E. Loeza-Kuk, O. Morales-Galvan, C. Chavarin-Palacio, J. Hinojosa-Moya, Localized expression of antimicrobial proteins mitigates huanglongbing symptoms in Mexican lime, *J. Biotechnol.* 285 (2018) 74–83.
- [54] J.C. Velasquez Guzman, S. Basu, R. Rabara, L.K. Huynh, G.C. Basu, H.B. Nguyen, Liposome delivery system of antimicrobial peptides against huanglongbing (HLB) citrus disease, *Biophys. J.* 114 (2018) 266a.
- [55] J.A. Qureshi, Dormant season foliar sprays of broad-spectrum insecticides: An effective component of integrated management for *Diaphorina citri* (Hemiptera: Psyllidae) in citrus orchards, *Crop Prot.* 29 (2010) 860–866.
- [56] D.R. Boina, Chemical control of the Asian citrus psyllid and of huanglongbing disease in citrus, *Pest Manag. Sci.* 71 (2015) 808–823.
- [57] J.A. Qureshi, B.C. Kostyk, Insecticidal suppression of Asian citrus psyllid *Diaphorina citri* (Hemiptera: Liviidae) vector of huanglongbing pathogens, *PLoS One* 9 (2014) E112331.
- [58] P.A. Stansly, H.A. Arevalo, J.A. Qureshi, M.M. Jones, K. Hendricks, P.D. Roberts, Vector control and foliar nutrition to maintain economic sustainability of bearing citrus in Florida groves affected by huanglongbing, *Pest Manag. Sci.* 70 (2014) 415–426.
- [59] C. Monzo, Thresholds for vector control and compatibility with beneficial fauna in citrus with high incidence of huanglongbing, *Acta Hort* 1065 (2015) 1137–1144.
- [60] J.A. Tansey, P. Vanaclocha, C. Monzo, M. Jones, Costs and benefits of insecticide and foliar nutrient applications to huanglongbing-infected citrus trees, *Pest Manag. Sci.* 73 (2017) 904–916.
- [61] C. Monzo, Economic injury levels for Asian citrus psyllid control in process oranges from mature trees with high incidence of huanglongbing, *PLoS One* 12 (2017) e0175333.
- [62] J. Belasque Jr., R.B. Bassanezi, P.T. Yamamoto, A.J. Ayres, A. Tachibana, A. R. Violante, A. Tank Jr., F. Di Giorgi, F.E.A. Tersi, Lessons from huanglongbing management in São Paulo state, Brazil, *J. Plant Pathol.* 92 (2010) 285–302.
- [63] J.A. Lopez, Huanglongbing and the California citrus industry: A cost comparison of do nothing vs. do something management practices, *Tex. J. Agr. Nat. Resour.* 27 (2015) 51–68.
- [64] X.D. Chen, T.A. Gill, M. Ashfaq, K.S. Pelz-Stelinski, Resistance to commonly used insecticides in Asian citrus psyllid: Stability and relationship to gene expression, *J. Appl. Entomol.* 142 (2018) 967–977.
- [65] M. Ozores-Hampton, F.M. Roka, R.E. Rouse, Dual treatment tested for HLB trees, *Citrus Industry News*, May (2017) 28–31.
- [66] R.E. Rouse, M. Ozores-Hampton, F.M. Roka, Rehabilitation of huanglongbing affected citrus trees using severe pruning and enhanced foliar nutritional treatments, *HortScience* 52 (2017) 972–978.
- [67] K.T. Morgan, R.E. Rouse, Foliar applications of essential nutrients on growth and yield of 'Valencia' sweet orange infected with huanglongbing, *HortScience* 51 (2016) 1482–1493.
- [68] C. Yang, C.A. Powell, Y. Duan, R.G. Shatters, Y. Lin, Mitigating citrus huanglongbing via effective application of antimicrobial compounds and thermotherapy, *Crop Prot.* 84 (2016) 150–158.
- [69] M.T. Hoffman, M.S. Doud, L. Williams, M.-Q. Zhang, F. Ding, E. Stover, D. Hall, S. Zhang, L. Jones, Heat treatment eliminates '*Candidatus Liberibacter asiaticus*' from infected citrus trees under controlled conditions, *Phytopathology* 103 (2013) 15–22.
- [70] Zhang, L., S. Gao, D. Xie, and F. Zhang. 2016. Varying pulse control schemes for citrus huanglongbing epidemic model with general incidence. *Commun. Math. Biol. Neurosci.* 2016: Article ID 7.
- [71] G. Fan, Y. Xia, X. Lin, H. Hu, X. Wang, C. Ruan, L. Lu, Evaluation of thermotherapy against huanglongbing (citrus greening) in the greenhouse, *J. Integr. Agr.* 15 (2016) 111–119.
- [72] R. Ehsani, Tenting and supplementary heat for thermotherapy of HLB trees, *Citrus Industry News*, Aug. (2015) 8–12.
- [73] A. Al-Jumaili, Mobile batch heat treatment system for treating HLB infected citrus trees. 2015 ASABE Annu. Intl. Meet. 1 (2015).
- [74] R. Ehsani, J.I.R. De Corcuera, The potential of thermotherapy in combatting HLB, *Resour. Mag.* 20 (2013) 18–19.
- [75] R. Ehsani, M. Dewdney, Controlling HLB with thermotherapy: What have we learned so far? *Citrus Industry News*, Sept. (2016) 26–28.
- [76] Dewdney, M., E. Johnson, N. Thapa, and M. Danyluk. 2018. Thermotherapy's effects on fruit drop, yield and quality. *Citrus Industry News*, Dec.: 12–15.
- [77] S. Li, F. Wu, Y. Duan, A. Singerman, Z. Guan, Citrus greening: Management strategies and their economic impact, *HortScience* 55 (5) (2020) 604–612.
- [78] S. Alvarez, E. Rohrig, D. Solís, Citrus greening disease (huanglongbing) in Florida: Economic impact, management and the potential for biological control, *Agr. Res.* 5 (2016) 109–118.
- [79] Muraro, R.P. 2010. Costs of managing HLB and citrus black spot. Presented at Citrus Expo, Ft. Meyers, FL. 19 May 2010.
- [80] F.R. Roka, A.M. Muraro, Economics of HLB management: Pull trees or spray nutritional, *Int. Citrus Econ. Conf.* (2010).
- [81] T.R. Gottwald, D.G. Hall, A.B. Kriss, E.J. Salinas, P.E. Parker, G.A.C. Beattie, Orchard and nursery dynamics of the effect of interplanting citrus with guava for huanglongbing, vector, and disease management, *Crop Prot.* 64 (2014) 93–103.
- [82] Z. Peng, J.V. Bredeson, G.A. Wu, S. Shu, N. Rawat, D. Du, S. Parajuli, Q. Yu, Q. You, D.S. Rokhsar, A chromosome-scale reference genome of trifoliate orange (*Poncirus trifoliata*) provides insights into disease resistance, cold tolerance and genome

- evolution in Citrus, *Plant J.* 104 (2020) 1215–1232, <https://doi.org/10.1111/tpj.14993>.
- [83] X. Zou, X. Jiang, L. Xu, T. Lei, A. Peng, Y. He, L. Yao, S. Chen, Transgenic citrus expressing synthesized *cecropin B* genes in the phloem exhibits decreased susceptibility to Huanglongbing, *Plant Mol. Biol.* 93 (2017) 341–353, <https://doi.org/10.1007/s11103-016-0565-5>.
- [84] C. Robertson, X. Zhang, S. Gowda, V. Orbovic, W.O. Dawson, Z. Mou, Overexpression of the Arabidopsis NPR1 protein in citrus confers tolerance to Huanglongbing, *J. Citrus Pathol.* 5 (2018) 1–8, <https://doi.org/10.5070/C451038911>.
- [85] H. Nguyen, G. Hao, E. Stover, G. Gupta, Novel therapy of highpriority Citrus diseases, *Citrograph* 7 (2016) 72–75.